

Algorithm Templates — Concise (Templates Only)

Canonical templates, core algorithms, cheat sheets, and pattern choosers only — no per-problem quick-reference tables and no problem index. See `template.md` for the full version with reference tables.

Two Pointers `two_pointers.md`

Two patterns — pick based on whether pointers converge or march together.

Pattern 1 — Opposite Two Pointers

```
// MENTAL MODEL: two ends close in on the answer; each comparison rules out one whole end of the range.
// WHY sorted: moving inward only safely eliminates half the space if the array is sorted
int i = 0, j = n - 1;
while (i < j) {
    if (condition == target) {
        /* record */ i++; j--;
    } else if (tooSmall) {
        i++;
    } else {
        j--;
    }
}
```

Sort first. `i` and `j` converge inward. Used when the array is sorted and you can eliminate half the search space each step.

Pattern 2 — Same Direction (Write Pointer)

```
// MENTAL MODEL: a fast reader scans everything; a slow writer trails behind and only copies the keeper
// WHEN: filter/compact an array in place, no extra buffer
int i = 0; // i = write position
for (int j = 0; j < n; j++) { // j = read position
    if (keepCondition(nums[j]))
        nums[i++] = nums[j];
}
return i;
```

`i` marks where the next valid element goes. `j` scans every element. Only write when the element passes the keep condition.

The generalised duplicate-skip pattern for "at most k duplicates":

```
int i = 0;
for (int j = 0; j < n; j++)
    if (i < k || nums[j] != nums[i - k]) // compare write head to k slots back
        nums[i++] = nums[j];
return i;
```

- $k = 1 \rightarrow$ #26 (no duplicates)
 - $k = 2 \rightarrow$ #80 (at most 2)
-

Pattern Comparison

Pattern	Pointers move	Sort needed	Use when
Opposite	<code>i</code> right, <code>j</code> left, converge	Yes (usually)	Sum/target problems on sorted array
Same direction	<code>j</code> scans, <code>i</code> writes	No	Filter / compact array in-place
Dutch Flag	<code>k</code> scans, <code>i</code> low boundary, <code>j</code> high boundary	No	3-way partition (< / = / >)

Duplicate-skip Cheat Sheet (for Opposite pointers after a hit)

```
// After recording a result at (i, j):  
i++; j--;  
while (i < j && nums[i] == nums[i - 1]) i++; // skip duplicate i  
while (i < j && nums[j] == nums[j + 1]) j--; // skip duplicate j
```

Always check `i < j` inside the skip loop to avoid crossing.

Sliding Window `sliding_window.md`

Three variants — pick by what the problem asks for.

`i` = left boundary (inclusive), `j` = right boundary (loop variable, inclusive).

Window = `[i, j]`, size = `j - i + 1`.

The Three Templates

```
// FIXED window of size k – shrink exactly once when full
// MENTAL MODEL: a constant-width window slides one step at a time; add the new cell, drop the old one.
// WHEN: "subarray/substring of size k"
int i = 0;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    if (j - i + 1 == k) {
        // 2. record (window is exactly size k)
        // 3. remove nums[i] from state
        i++;
    }
}

// MAX variable window – find LONGEST valid window
// MENTAL MODEL: grow greedily; shrink only enough to restore validity, then the window is the best end
// WHEN: "longest" + "at most k ..."
int i = 0, result = 0;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    while (/* window exceeds constraint */) { // shrink WHILE window exceeds constraint
        // remove nums[i] from state
        i++;
    }
    result = Math.max(result, j - i + 1); // record after window is valid
}

// MIN variable window – find SHORTEST valid window
// MENTAL MODEL: grow until valid, then shrink aggressively while still valid to find the tightest fit.
// WHEN: "shortest/minimum" + "sum >= target" / "contains all of t"
int i = 0, result = Integer.MAX_VALUE;
for (int j = 0; j < n; j++) {
    // 1. add nums[j] to state
    while (/* window VALID */) { // record then shrink WHILE window still valid
        result = Math.min(result, j - i + 1); // record while valid
        // remove nums[i] from state
        i++; // then try to shrink
    }
}
```

Frequency Map Trick (used by 567 and 76)

Both use a `need[]` array and a `required` counter to avoid scanning the map each step.

```
// Expanding j – add s.charAt(j):
if (need[s.charAt(j)]-- > 0) required--; // was needed → satisfied one more

// Shrinking i – remove s.charAt(i):
if (need[s.charAt(i)]++ >= 0) required++; // was 0 (used up) → now short one
i++;
```

`need[c]` is positive when still needed, zero or negative when excess.

`required` reaches 0 exactly when all characters of `t` are covered.

Choosing the Template

Question asks for	Template	i moves
Fixed-size window operation	Fixed	when <code>j-i+1 == k</code>
Longest / maximum window	Max variable	while invalid (<code>while</code>)
Shortest / minimum window	Min variable	while valid (<code>while</code>)

Window content	State to maintain
Sum of values	<code>int sum</code>
Unique characters / no duplicates	<code>int[] freq</code> (128 or 26) or <code>Set</code>
Character frequency match	<code>int[] need</code> + <code>int required</code> counter
Running maximum	<code>Deque<Integer></code> monotone decreasing indices

Prefix Sum + HashMap `prefix_sum_hashmap.md`

Core idea: `subarray sum [l, r] = prefix[r+1] - prefix[l]`.

To find a subarray with a target property, rephrase as: "has a prefix with value X been seen before?"

Store that X in a HashMap for O(1) lookup.

Core Template

```
// MENTAL MODEL: a subarray is the difference of two prefixes; remember prefixes seen so far and look u
// WHEN: "subarray sum equals/divisible by k", "longest subarray with sum X" – anything reducible to pr
Map<Long, Integer> map = new HashMap<>();
map.put(0L, ???); // seed: empty prefix (sum=0) seen before index 0
long sum = 0;
// result = 0 (count) or Integer.MAX_VALUE (length sentinel) depending on problem

for (int i = 0; i < nums.length; i++) {
    sum += nums[i];
    long lookup = transform(sum); // what to look up (sum-k, sum%k, etc.)
    // use map.get(lookup) → update result
    map.put(storeKey(sum), storeValue(i)); // what to store (sum or sum%k → count or index)
}
```

What changes per problem:

Variable	Count problems (560, 974)	Length problems (325, 523)
Map value	<code>count</code> (how many times seen)	<code>first index</code> (earliest occurrence)
Seed value	<code>map.put(0L, 1)</code>	<code>map.put(0L, -1)</code>
On lookup hit	<code>count += map.get(lookup)</code>	<code>maxLen = max(i - map.get(lookup))</code>
Overwrite map?	Yes (merge count)	No (keep first index only)

When to Use `long` vs `int`

- Use `long` for the prefix sum when values can be large (e.g., `nums[i]` up to $10^4 \times n$ up to $2 \times 10^4 \rightarrow$ sum up to 2×10^8 , safe as int, but $10^5 \times 10^5 = 10^{10}$ overflows).
- Use `int` for the remainder key (modulo result is always in `[0, k-1]`).

Why 0L (not 0) in the seed

When the map is `Map<Long, Integer>`, the seed **key** must be a `long` literal:

```
Map<Long, Integer> map = new HashMap<>();
map.put(0L, 1);      // ✓ long literal → boxes to Long (matches key type)
map.put(0, 1);       // ✗ compile error: int boxes to Integer, not Long
```

- The `L` suffix makes it a `long` literal so it autoboxes to `Long`, matching the key type (the keys are `long` because `sum` is `long` for the overflow guard above).
- **Subtle trap:** even numerically equal boxes of different types never compare equal — `Long.valueOf(0).equals(Integer.valueOf(0))` is `false`. Keeping every key `long` / `Long` avoids silent lookup misses.
- If `int` sums can't overflow, use `Map<Integer, Integer>` with plain `map.put(0, 1)` instead — simpler, no `L` needed.

Binary Search `binary_search.md`

Binary search appears in two forms: searching an **array index**, or searching an **answer value**.

The Only Template — `[i, j)`, find first `k` with `condition(k)`

There is one binary search. Define a monotone `condition(k)` (`false...false TRUE...true`); the loop returns the **first `k` where `condition(k)` is true**. Then look at `i`: if `i == n` no index qualified, otherwise `i` is the boundary — check the value to decide.

```
int i = 0, j = nums.length;      // [i, j)
// find first k that meets condition(k)
while (i < j) {
    int k = i + (j - i) / 2;
    if (!condition(k)) {          // condition false → answer is strictly to the right
        i = k + 1;
    } else {                      // condition true → k might be the answer
        j = k;
    }
}
// i in [0, nums.length]; if i == nums.length, no k met condition(k)
```

Pick `condition(k)`:

Want	condition(k)	Answer after loop
lowerBound — first <code>>= target</code>	<code>nums[k] >= target</code>	<code>i</code>
upperBound — first <code>> target</code>	<code>nums[k] > target</code>	<code>i</code>
exact search (#704)	<code>nums[k] >= target</code>	<code>i < n && nums[i] == target ? i : -1</code>
insert position (#35)	<code>nums[k] >= target</code>	<code>i</code>
first / last occurrence (#34)	<code>>=</code> then <code>></code>	<code>lowerBound, upperBound - 1</code>
count of target	—	<code>upperBound - lowerBound</code>
minimize feasible X	<code>feasible(k)</code> over value range	<code>i</code>
maximize feasible X	<code>!feasible(k)</code> (first infeasible)	<code>i - 1</code>

The rule: the answer is always `i`. Guard `i == nums.length` (nothing met the condition) before reading `nums[i]`, then confirm the value.

How to Choose

Signal	condition(k)
Sorted array, find exact value	<code>arr[k] >= target</code> , then check <code>arr[i] == target</code>
Insert position / first occurrence	<code>arr[k] >= target</code> → return <code>i</code> (= <code>lowerBound</code>)
Last occurrence / count	<code>arr[k] > target</code> (= <code>upperBound</code>); <code>last = upperBound - 1</code> , <code>count = upper - lower</code>
Rotated / peak slope search	<code>condition(k)</code> compares <code>arr[k]</code> to a neighbor or endpoint
"minimum X such that feasible(X)"	<code>condition(k) = feasible(k)</code> over value range → return <code>i</code>
"maximum X such that feasible(X)"	<code>condition(k) = !feasible(k)</code> over <code>[lo, hi+1)</code> → return <code>i - 1</code>

The one rule: define `condition(k)` so it is monotone `false...false TRUE...true` , then the answer is the first `k` that flips it true (`j = k` on true, `i = k + 1` on false). For MAXIMIZE, make the condition "first infeasible" and subtract 1 — same loop, no ceiling mid.

Search on the **answer value** itself, not an array index. All use half-open `[i, j)` with `while (i < j)` .

Sorting `sorting.md`

All recursive sorting uses **half-open intervals** `[start, end)` throughout.
Inside merge and partition: `i` , `j` , `k` as scan/write pointers (consistent with binary search convention).

Template 1 — Merge Sort

```
// MENTAL MODEL: split in half until trivially sorted, then merge two sorted halves into one.
// WHEN: need stable sort, sort a linked list, or count cross-pairs (inversions) during the merge.
// Entry point
public void mergeSort(int[] nums) {
    mergeSort(nums, 0, nums.length, new int[nums.length]);
}

// [start, end) half-open
private void mergeSort(int[] nums, int start, int end, int[] temp) {
    if (end - start <= 1) return; // base case: end - start <= 1 → 0 or 1 elements, already sort
    int mid = start + (end - start) / 2;
    mergeSort(nums, start, mid, temp); // sort [start, mid)
    mergeSort(nums, mid, end, temp); // sort [mid, end)
    merge(nums, start, mid, end, temp);
}

// i scans left [start,mid), j scans right [mid,end), k writes to temp
private void merge(int[] nums, int start, int mid, int end, int[] temp) {
    int i = start, j = mid, k = start;
    while (i < mid || j < end) {
        if (j >= end || (i < mid && nums[i] <= nums[j])) {
            temp[k++] = nums[i++];
        } else {
            temp[k++] = nums[j++];
        }
    }
    for (i = start; i < end; i++) nums[i] = temp[i];
}
```

Template 2 — Quick Sort (3-way Dutch Flag Partition)

```
// MENTAL MODEL: pick a pivot, partition into <pivot | ==pivot | >pivot, recurse on the two ends only.
// WHEN: in-place sort with O(log n) stack; the ==pivot middle is already in final position.
// Entry point – no extra array needed (in-place)
public void quickSort(int[] nums) {
    quickSort(nums, 0, nums.length);
}

// [start, end) half-open
private void quickSort(int[] nums, int start, int end) {
    if (end - start <= 1) return;
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    // Invariant: [start,i) < pivot | [i,k) == pivot | [k,j) unknown | (j,end) > pivot
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    // After: [start,i) < pivot, [i,k) == pivot, [k,end) > pivot
    quickSort(nums, start, i); // recurse on < region
    quickSort(nums, k, end); // recurse on > region
}

private void swap(int[] nums, int i, int j) {
    int t = nums[i]; nums[i] = nums[j]; nums[j] = t;
}
```

Template 3 — Quick Select (partial sort — k-th element)

Extends quick sort: after partition, only recurse into the side that contains `target` index.

```
// MENTAL MODEL: same partition as quick sort, but recurse into only the one side holding target → O(n)
// WHEN: "k-th largest/smallest", "k closest/most frequent" – you need a position, not a full sort.
// Returns value at 0-indexed position target after sorting
private int quickSelect(int[] nums, int start, int end, int target) {
    int pivot = nums[start + (end - start) / 2];
    int i = start, k = start, j = end - 1;
    while (k <= j) {
        if (nums[k] < pivot) {
            swap(nums, k++, i++);
        } else if (nums[k] == pivot) {
            k++;
        } else {
            swap(nums, k, j--);
        }
    }
    // [start,i) < pivot, [i,k) == pivot, [k,end) > pivot
    if (target < i) {
        return quickSelect(nums, start, i, target);
    } else if (target < k) {
        return pivot; // target lands in == region
    } else {
        return quickSelect(nums, k, end, target);
    }
}
```

Template 4 — Counting Sort

$O(n + \text{range})$. Use when values are bounded integers.

```
// MENTAL MODEL: tally how many times each value occurs, then emit values in order — no comparisons.
// WHEN: integers in a small bounded range; beats  $O(n \log n)$  when  $\text{range} = O(n)$ .
public void countingSort(int[] arr) {
    if (arr.length == 0) return;
    int min = Arrays.stream(arr).min().getAsInt();
    int max = Arrays.stream(arr).max().getAsInt();
    int[] buckets = new int[max - min + 1];
    for (int x : arr) buckets[x - min]++;
    int i = 0;
    for (int v = 0; v < buckets.length; v++)
        while (buckets[v]-- > 0) arr[i++] = v + min;
}
```

Algorithm Chooser

Signal	Algorithm
"sort array", stable, $O(n)$ space OK	Merge Sort
"sort array", in-place, $O(\log n)$ space	Quick Sort
"k-th largest/smallest"	Quick Select — $O(n)$ avg
"k closest / k most frequent"	Quick Select with custom comparator
"sort linked list"	Merge Sort (quick sort needs random access)
"count inversions / smaller after self"	Merge Sort — count during merge step
"arrange to maximize/compare"	Custom Sort comparator
"merge overlapping intervals"	Sort by start + linear scan
integers in bounded range	Counting Sort — $O(n + \text{range})$

Half-open Interval Cheat Sheet

```
mergeSort(nums, start, end) → sorts [start, end)
    mid = start + (end-start)/2
    left:  [start, mid)
    right: [mid, end)
    base:  end - start <= 1

quickSort(nums, start, end) → sorts [start, end)
    after partition: [start,i) < pivot, [i,k) == pivot, [k,end) > pivot
    recurse: (start, i) and (k, end)
```

Linked List `linked_list.md`

Four patterns cover almost every linked list problem.

Consistent naming throughout: `sentinel`, `previous`, `current`, `next`, `slow`, `fast`.

When to Use — Signal → Pattern

Signal in the prompt	Pattern
"remove / delete / skip" nodes by value or duplicate	Delete / Filter (sentinel + previous)
"reverse" the list, a sub-range, or in k-groups	Reversal
"middle node", "cycle", "nth node from the end"	Fast / Slow
"merge two sorted lists" (or k lists)	Merge (sentinel + heap for k)
chained steps ("reorder", "palindrome", "sort")	Composite (combine the above)

All Four Templates

```
// 1. DELETE / FILTER – sentinel guards head removal; previous tracks last kept node
// MENTAL MODEL: previous always points at the last node you decided to keep, so re-wiring it skips any
// WHEN: "remove / delete / skip nodes by value or duplicate"
ListNode sentinel = new ListNode(0);
sentinel.next = head;
ListNode previous = sentinel, current = head;
while (current != null) {
    if (shouldDelete(current)) {
        previous.next = current.next;    // skip node
    } else {
        previous = current;              // keep node, advance previous
    }
    current = current.next;
}
return sentinel.next;

// 2. REVERSAL – re-wire one node at a time
// MENTAL MODEL: flip each arrow to point backward; previous is the growing reversed list trailing behind
// WHEN: "reverse the list (whole, partial, or in k-groups)"
ListNode previous = null, current = head;
while (current != null) {
    ListNode next = current.next;    // save before overwrite
    current.next = previous;         // reverse pointer
    previous = current;              // advance
    current = next;
}
return previous; // new head

// 3. FAST / SLOW – two pointers at different speeds
// MENTAL MODEL: fast covers twice the distance, so when it hits the end slow is exactly halfway; in a
// WHEN: "middle node", "detect/locate cycle", "nth from end"
ListNode slow = head, fast = head;
while (fast != null && fast.next != null) {
    slow = slow.next;
    fast = fast.next.next;
}
// slow is at middle (or cycle detection: slow == fast → cycle found)

// 4. MERGE – sentinel collects nodes from two lists in order
// MENTAL MODEL: repeatedly splice the smaller head onto a growing result tail, like a zipper.
// WHEN: "merge two (or k) sorted lists"
ListNode sentinel = new ListNode(0);
ListNode current = sentinel;
while (a != null && b != null) {
    if (a.val <= b.val) {
        current.next = a;
        a = a.next;
    } else {
        current.next = b;
        b = b.next;
    }
    current = current.next;
}
current.next = (a != null) ? a : b;
return sentinel.next;
```

`previous` stays on the last **kept** node. When deleting, re-wire `previous.next` to skip `current`. When keeping, advance `previous` to `current`. Always advance `current`.

`fast` moves 2× per step. When `fast` reaches the end, `slow` is at the middle.

When `slow == fast` after the start, a cycle is detected.

`sentinel` collects the merged result. `current` is the write pointer.

Problems that chain multiple patterns together.

Pattern Summary

Pattern	Sentinel	Naming	When to use
Delete / Filter	Yes	<code>previous</code> , <code>current</code>	Remove or skip nodes by condition
Reversal	No (Yes for partial)	<code>previous</code> , <code>current</code> , <code>next</code>	Re-wire pointer direction
Fast / Slow	No	<code>slow</code> , <code>fast</code>	Middle, cycle, nth from end
Merge	Yes	<code>current</code>	Combine two or more lists
Composite	Both	Mix of above	Reorder, sort, rearrange

Sentinel Cheat Sheet

SENTINEL: dummy node before head so head removal needs no special-casing; use it whenever the head itself might be removed/replaced.

```
// Always use sentinel when the head itself might be removed or replaced:
ListNode sentinel = new ListNode(0);
sentinel.next = head;
// ... operations ...
return sentinel.next;    // head may have changed; sentinel.next is always correct
```

String `string.md`

Core idioms first, then problems grouped by pattern.

For sliding window string problems (#3, #76, #567, #424), see `sliding_window.md` .

Core Idioms

```
// MENTAL MODEL: a lowercase string is a length-26 frequency vector; most string problems are vector op
// WHEN: "anagram", "char count", "group by letters", "first unique", "sort by frequency".
// 1. CHAR COUNT – lowercase letters
int[] count = new int[26];
// WHY ch - 'a': ASCII 'a'=97, so 'a'→0, 'b'→1, ..., 'z'→25 – a clean 0–25 bucket index into count[].
for (char ch : s.toCharArray()) count[ch - 'a']++; // ← ch - 'a' maps a→0, b→1, ..., z→25

// 2. CHAR COUNT – digits 0–9
int[] count = new int[10];
for (char ch : s.toCharArray()) count[ch - '0']++; // ← ch - '0' maps '0'→0, '9'→9

// 3. CHAR TO INTEGER – build number digit by digit
int num = 0;
for (int i = 0; i < s.length(); i++)
    num = num * 10 + (s.charAt(i) - '0'); // ← shift left × 10, add new digit

// 4. ANAGRAM HASH KEY – frequency-based (bucket sort style) O(k) vs sort O(k log k)
// WHY it works: anagrams share identical letter frequencies, so encoding the frequency vector
// ITSELF gives a unique key for the whole group – O(k) to build vs O(k log k) to sort the chars.
String hashKey(String s) {
    int[] count = new int[26];
    for (char ch : s.toCharArray()) count[ch - 'a']++;
    StringBuilder builder = new StringBuilder();
    for (int i = 0; i < 26; i++) {
        builder.append((char)('a' + i)); // ← letter as label
        builder.append(count[i]);       // ← its count
    }
    return builder.toString(); // e.g., "a2b0c1...z0"
}
// Alternative: sort the characters (simpler, slightly slower)
char[] chars = s.toCharArray();
Arrays.sort(chars);
String key = new String(chars); // anagrams → same sorted key

// 5. BUCKET SORT – sort by frequency without comparison sort
int[] count = new int[128]; // ASCII
for (char ch : s.toCharArray()) count[ch]++;
List<Character>[] buckets = new List[s.length() + 1]; // index = frequency
for (int i = 0; i < 128; i++) {
    if (count[i] > 0) {
        int freq = count[i];
        if (buckets[freq] == null) buckets[freq] = new ArrayList<>();
        buckets[freq].add((char) i);
    }
}
StringBuilder builder = new StringBuilder();
for (int freq = buckets.length - 1; freq > 0; freq--) { // ← descending frequency
    if (buckets[freq] == null) continue;
    for (char ch : buckets[freq])
        for (int i = 0; i < freq; i++) builder.append(ch);
}

// 6. EXPAND FROM CENTER – palindrome check in O(1) space
int expand(String s, int i, int j) { // i == j: odd length; i+1 == j: even length
    while (i >= 0 && j < s.length() && s.charAt(i) == s.charAt(j)) { i--; j++; }
    return j - i - 1; // ← length of palindrome found
}
```

Template

```
// Call for each center: odd-length palindromes (i == i) and even-length (i, i+1)
for (int i = 0; i < s.length(); i++) {
    // odd: single center at i
    // even: center between i and i+1
    processExpansion(s, i, i);    // odd
    processExpansion(s, i, i + 1); // even
}

// Shared expand helper – returns length of palindrome
int expand(String s, int i, int j) {
    while (i >= 0 && j < s.length() && s.charAt(i) == s.charAt(j)) { i--; j++; }
    return j - i - 1;
}
```

String Idioms Cheat Sheet

```
// Char ↔ index
'a' + i           // int → char (i=0→'a', i=25→'z')
ch - 'a'          // char → index (a→0, z→25)
ch - '0'          // char → digit (0→0, 9→9)
Character.isDigit(ch)
Character.isLetter(ch)
Character.toLowerCase(ch)

// String operations
s.charAt(i)                // char at index
s.toCharArray()           // → char[]
new String(charArray)      // char[] → String
String.valueOf(charArray)  // char[] → String
s.substring(i, j)          // [i, j)
s.contains(sub)            // boolean
s.startsWith(prefix)       // boolean
s.split("\\s+")            // split on whitespace

// Building strings
StringBuilder builder = new StringBuilder();
builder.append(ch);         builder.append(str);
builder.deleteCharAt(i);
builder.reverse();
builder.toString();
String.join(", ", list)     // join with delimiter

// computeIfAbsent – key pattern for grouping
map.computeIfAbsent(key, k -> new ArrayList<>()).add(value);
```

Pattern Chooser

Signal	Pattern
Are two strings anagrams?	Count array: fill from s, drain from t
Group strings by anagram	Frequency hash key (<code>hashCode</code>) or <code>Arrays.sort</code> key
Sort/rank by character frequency	Bucket sort: <code>count[]</code> , then <code>buckets[freq]</code>
Longest/count palindromic substrings	Expand from center (i,i) odd + (i,i+1) even
Parse a number from string	<code>num = num * 10 + (ch - '0')</code>
Remove adjacent duplicate pairs	Stack: push, pop on match
Sliding window over characters	See <code>sliding_window.md</code>

Stack / Queue / Heap `stack_queue.md`

Four data structures — each with a canonical template and representative problems.

All Templates

```
// 1. STACK (LIFO) – use ArrayDeque, not Stack class
Deque<Integer> stack = new ArrayDeque<>();
stack.push(x);      // add to top
stack.pop();        // remove from top
stack.peek();       // view top, no remove

// 2A. MONOTONE DECREASING STACK – next GREATER element
// MENTAL MODEL: keep only candidates still waiting for a bigger neighbor; current resolves them all.
// WHEN: "next/previous greater element", "warmer/taller to the right"
// Invariant: stack values decrease from bottom to top
// Pop when current is GREATER than top → top found its next greater
Deque<Integer> stack = new ArrayDeque<>(); // stores indices
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] < nums[i])
        result[stack.pop()] = nums[i]; // nums[i] is next greater for popped index
    stack.push(i);
}
// Remaining in stack: no next greater element exists

// 2B. MONOTONE INCREASING STACK – next SMALLER element / span width
// MENTAL MODEL: each popped bar's reach extends until something shorter blocks it on both sides.
// WHEN: "largest rectangle", "trapped water", "span/width bounded by smaller elements"
// Invariant: stack values increase from bottom to top
// Pop when current is SMALLER than top → use popped index to compute width/span
Deque<Integer> stack = new ArrayDeque<>(); // stores indices
for (int i = 0; i < n; i++) {
    while (!stack.isEmpty() && nums[stack.peek()] > nums[i]) {
        int height = nums[stack.pop()];
        int width = stack.isEmpty() ? i : i - stack.peek() - 1; // span between boundaries
        // process height * width
    }
    stack.push(i);
}

// 3. MONOTONE DEQUE – sliding window max/min
// MENTAL MODEL: the front is the current window's answer; anything a newer-and-bigger value beats is d
// WHEN: "max/min of every window of size k"
// Deque stores INDICES; values at those indices are decreasing front-to-back (for max)
Deque<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) {
    while (!queue.isEmpty() && nums[queue.peekLast()] <= nums[i])
        queue.pollLast(); // remove smaller elements – they can never be window max
    queue.offerLast(i);
    if (queue.peekFirst() < i - k + 1) queue.pollFirst(); // evict out-of-window elements
    if (i >= k - 1) result[i - k + 1] = nums[queue.peekFirst()];
}

// 4. PRIORITY QUEUE
PriorityQueue<Integer> minPQ = new PriorityQueue<>(); // min-heap (default)
PriorityQueue<Integer> maxPQ = new PriorityQueue<>(Collections.reverseOrder()); // max-heap
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]); // custom: sort by index 1
pq.offer(x); // add
pq.poll(); // remove and return min/max
pq.peek(); // view min/max without removing

// 5. TWO HEAPS – maintain median in O(log n)
// MENTAL MODEL: split the data at the median; the median is always sitting on the two heaps' tops.
// WHEN: "running median", "median of a data stream"
// maxHeap = lower half (smaller numbers), minHeap = upper half (larger numbers)
// Invariant: maxHeap.size() == minHeap.size() or maxHeap.size() == minHeap.size() + 1
PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder());
```

```

PriorityQueue<Integer> minHeap = new PriorityQueue<>();
void addNum(int num) {
    maxHeap.offer(num);
    minHeap.offer(maxHeap.poll());           // push one to minHeap (possibly num itself)
    if (maxHeap.size() < minHeap.size())
        maxHeap.offer(minHeap.poll());       // rebalance: maxHeap always ≥ minHeap in size
}
double findMedian() {
    return maxHeap.size() > minHeap.size()
        ? maxHeap.peek()
        : (maxHeap.peek() + minHeap.peek()) / 2.0;
}

```

Decreasing vs Increasing

DECREASING stack (pop when current > top): finds NEXT GREATER element for each popped index
 INCREASING stack (pop when current < top): finds NEXT SMALLER element – used for span width

Both store INDICES (not values) so you can compute distances and widths.

Structure Chooser

Goal	Structure	Why
Next greater/smaller element	Monotone stack	O(n) total — each element pushed/popped once
Sliding window max/min	Monotone deque	Front = window extreme; stale indices evicted
Global min/max, k-th element	PriorityQueue	O(log n) per op
Running median	Two heaps	Split at median; each heap half is O(log n)
O(1) stack minimum	Aux min-stack	Mirror min value at every level
Nested structure (brackets, k-repeats)	Two stacks	One for counts, one for strings

Deque API Cheat Sheet

One class, three roles. `ArrayDeque` adds/removes at *both* ends, so stack, queue, and deque are just usage patterns of the same structure — the only difference is **which end you remove from**. Always declare it `Deque<Integer> queue = new ArrayDeque<>();` (never the legacy `Stack` class, which is synchronized and `Vector`-backed; and `ArrayDeque` beats `LinkedList` as a queue). Name the variable for the *role* it plays (`stack` / `queue`).

addFirst / push	addLast / offer
↓	↓
[FRONT BACK]	
↑	↑
pollFirst / pop	pollLast

STACK (LIFO): add + remove at FRONT → newest out first

QUEUE (FIFO): add at BACK, remove at FRONT → oldest out first

Intent	Stack (LIFO)	Queue (FIFO)
Add	<code>push(x)</code> → <code>addFirst</code>	<code>offer(x)</code> → <code>addLast</code>
Remove	<code>pop()</code> → <code>removeFirst</code>	<code>poll()</code> → <code>removeFirst</code>
Peek	<code>peek()</code> → <code>peekFirst</code>	<code>peek()</code> → <code>peekFirst</code>

Both **remove from the front** — the discipline comes from *where you add*: stack adds to the front (so newest is popped = LIFO), queue adds to the back (so oldest is polled = FIFO).

```
Deque<Integer> queue = new ArrayDeque<>();

// As STACK (LIFO):    push()/pop()/peek() → operate on FRONT
queue.push(x);        // = offerFirst
queue.pop();           // = pollFirst
queue.peek();          // = peekFirst

// As QUEUE (FIFO):    offer()/poll()/peek() → back-in, front-out
queue.offerLast(x);
queue.pollFirst();
queue.peekFirst();

// As DEQUE:           explicit first/last (e.g. monotone sliding-window)
queue.offerFirst(x);   queue.pollFirst(); queue.peekFirst();
queue.offerLast(x);    queue.pollLast();  queue.peekLast();
```

BFS (Tree) `bfs.md`

Core structure: `Queue<TreeNode> queue = new ArrayDeque<>()`.

The only decision: do you need to know **which level** a node is on?

The Two Templates

```
// 1. LEVEL-AWARE PROCESSING — snapshot size to make per-level decisions
// MENTAL MODEL: the queue holds exactly one full level; freeze its size, then drain just that many.
// WHEN: "level by level", "per row", "rightmost/leftmost in level", "min depth"
Queue<TreeNode> queue = new ArrayDeque<>();
queue.offer(root);
int level = 0;
while (!queue.isEmpty()) {
    int size = queue.size();          // ← snapshot: all nodes currently in queue = this level
    level++;
    for (int i = 0; i < size; i++) {
        TreeNode node = queue.poll();
        // process node (know: level, i==0 → first, i==size-1 → last)
        if (node.left != null) queue.offer(node.left);
        if (node.right != null) queue.offer(node.right);
    }
}

// 2. WITHOUT LEVEL TRACKING — simple node-by-node
// MENTAL MODEL: just visit every node in BFS order; you don't care which level it's on.
Queue<TreeNode> queue = new ArrayDeque<>();
queue.offer(root);
while (!queue.isEmpty()) {
    TreeNode node = queue.poll();
    // process node
    if (node.left != null) queue.offer(node.left);
    if (node.right != null) queue.offer(node.right);
}
```

Almost all tree BFS problems use **Template 1** — you nearly always need `size` to know when a level ends.

When to Use — Signal → Pattern

Problem signal phrase	Pattern / group
"level order", "values per level", "each row"	Group 1 — collect entire level
"average / max / sum of each level"	Group 2 — aggregate per level
"right side view" (last in level), "bottom-left" (first in level)	Group 2 — record boundary node (<code>i==size-1</code> / <code>i==0</code>)
"minimum depth", "first level that satisfies X"	Group 3 — early return on condition
"connect next pointers at the same level"	Group 4 — wire nodes within a level
"same depth, different parent" (cousins), per-node metadata	Group 5 — track per-node metadata
"maximum width of a level" (counting null gaps)	Group 6 — position indexing
no level info needed, just visit every node	Template 2 — node-by-node

The One Rule — Snapshot Level Size First

Always snapshot the level size BEFORE the inner for loop:

```
int size = queue.size();
for (int i = 0; i < size; i++) { ... }
```

Without the snapshot, newly added children mix with the current level and `i == size-1` / `i == 0` checks become meaningless.

Pattern Summary

What changes per level	Key variable	Example problems
Collect all values	<code>List<Integer> level</code>	102, 107, 103
Compute aggregate	<code>int sum</code> / <code>int max</code>	637, 515, 1161
Record boundary node	check <code>i == 0</code> or <code>i == size-1</code>	199, 513
Return on first match	<code>return depth</code>	111
Wire nodes together	<code>Node previous</code>	116, 117
Track parent metadata	<code>TreeNode xParent, yParent</code>	993

The One Rule (recap)

Snapshot the level size **before** the inner `for` loop — see "The One Rule" at the top of this file for the full explanation.

DFS / Backtracking `dfs.md`

DFS appears in five forms. The pattern determines naming, return type, and where the answer is built.

When to Use — Signal → Pattern

Problem signal phrase	Pattern / template
"height / depth / sum OF the subtree", combine results upward	#1 Process children first, combine at node (post-order)
"root-to-leaf path", "number built along the path"	#2 Process node first, pass state down (pre-order)
"longest/best path that may bend through a node" (spans both arms)	#3 Tree global answer
"count islands / largest region / flood fill" on a grid	#4 Grid DFS
"can finish all", "detect cycle", "valid course order" (directed)	#5 Graph DFS 3-color
"all subsets / permutations / combinations", "find every way"	#6 Backtracking
"valid BST", "kth smallest", "closest value", "successor"	BST templates (pass bounds / inorder / binary search)

All Templates

```
// 1. PROCESS CHILDREN FIRST, COMBINE AT NODE (post-order)
// MENTAL MODEL: each node trusts its children to return a finished answer, then merges them.
// WHEN: "what is the height/sum/property OF the subtree rooted here"
private int dfs(TreeNode node) {
    if (node == null) return BASE;           // 0 for depth, true for balanced
    int left = dfs(node.left);
    int right = dfs(node.right);
    // combine left, right, node.val → answer propagates up
    return ...;
}

// 2. PROCESS NODE FIRST, PASS STATE DOWN (pre-order)
// MENTAL MODEL: carry what you've seen so far down the path; the leaf has the full picture.
// WHEN: "root-to-leaf path", "running sum/number built along the way"
private void dfs(TreeNode node, int accumulated) {
    if (node == null) return;
    accumulated += node.val;                 // update state on way down
    if (isLeaf(node)) { /* record */ return; }
    dfs(node.left, accumulated);
    dfs(node.right, accumulated);
    // backtrack: no undo needed for primitive; undo List.add if using collection
}

// 3. TREE – GLOBAL ANSWER (return up the best single arm; update global across both arms)
// MENTAL MODEL: the answer may bend through a node using both arms, but only one arm can extend upward
// WHEN: use when the best path spans across a node, not just up one arm.
int answer = Integer.MIN_VALUE;
private int dfs(TreeNode node) {
    if (node == null) return 0;
    int left = Math.max(0, dfs(node.left)); // clamp negative
    int right = Math.max(0, dfs(node.right));
    answer = Math.max(answer, left + right + node.val); // cross-node path
    return Math.max(left, right) + node.val;           // single arm up
}

// 4. GRID DFS – mark visited by mutating grid; 4-directional flood fill
// MENTAL MODEL: stand on a cell, flood into all 4 neighbors, sinking each as you visit it.
// WHEN: "connected region / island / flood fill" on a grid
private void dfs(int[][] grid, int r, int c) {
    if (r < 0 || r >= grid.length || c < 0 || c >= grid[0].length) return;
    if (grid[r][c] != TARGET) return;
    grid[r][c] = VISITED;
    dfs(grid, r+1, c); dfs(grid, r-1, c);
    dfs(grid, r, c+1); dfs(grid, r, c-1);
}

// 5. GRAPH DFS – 3-color visited: 0=unseen 1=in-stack 2=done
// MENTAL MODEL: if you re-enter a node still on the current stack, you've looped back → cycle.
// WHEN: "can finish all", "detect cycle", "valid ordering" on a directed graph
int[] state;
private boolean dfs(int node) {
    if (state[node] == 1) return true;      // back edge → cycle
    if (state[node] == 2) return false;    // already safe
    state[node] = 1;
    for (int neighbor : graph.get(node))
        if (dfs(neighbor)) return true;
    state[node] = 2;
    return false;
}

// 6. BACKTRACKING – choose / recurse / undo
```

```
// MENTAL MODEL: build a candidate one element at a time; undo each choice to explore the next branch.
// WHEN: "all subsets / permutations / combinations", "find all ways"
private void backtrack(int start, List<Integer> current) {
    if (baseCase) { result.add(new ArrayList<>(current)); return; }
    for (int i = start; i < n; i++) {
        if (skipCondition(i)) continue; // prune duplicates
        current.add(nums[i]);           // choose
        backtrack(next, current);       // next = i (reuse) or i+1 (no reuse)
        current.remove(current.size()-1); // undo
    }
}
```

PROCESS CHILDREN FIRST, COMBINE AT NODE (post-order)

Recurse into children first, then combine results at the current node. Answer propagates upward.

PROCESS NODE FIRST, PASS STATE DOWN (pre-order)

State is built on the way **down** and consumed at leaves. Use backtracking when state is a mutable collection.

Grid DFS (4-directional flood fill)

Mark visited by overwriting the cell value. Return the accumulated property (count, area) or void.

Graph DFS — 3-Color Cycle Detection

Three states: `0` = unvisited, `1` = in current DFS stack (visiting), `2` = fully processed (safe).

A back edge (reaching a node with state `1`) means a cycle.

Template: choose → recurse → undo.

`start` controls which elements are eligible at each level. `current` is the in-progress candidate.

```
// MENTAL MODEL: grow one candidate by picking an element, recurse, then undo to try the next branch.
private void backtrack(int start, List<Integer> current) {
    if (baseCase) { result.add(new ArrayList<>(current)); return; }
    for (int i = start; i < n; i++) {
        if (skipCondition) continue;
        current.add(nums[i]);           // choose
        backtrack(nextStart, current); // nextStart = i (reuse) or i+1 (no reuse)
        current.remove(current.size() - 1); // undo
    }
}
```

Backtracking Decision Table

#	All orderings?	Reuse element?	Has duplicates?	Start index	Skip condition
46 Permutations	Yes — <code>used[]</code>	No	No	Loop 0..n, <code>used[i]</code>	<code>used[i]</code>
78 Subsets	No	No	No	<code>start → i+1</code>	None
39 Combination Sum	No	Yes	No	<code>start → i</code>	<code>candidates[i] > remaining</code>
40 Combination Sum II	No	No	Yes	<code>start → i+1</code>	<code>i > start && nums[i] == nums[i-1]</code>
131 Palindrome Part.	No	No	No	<code>start → end</code>	<code>!isPalindrome(sub)</code>

Tree DFS Summary

Pattern	When to use	Answer location
Return up	Combine children at node	return value bubbles up
Pass down	Accumulate state toward leaves	check at leaf
Global variable	Path spans across a node (not up)	int/long field, updated inside dfs

BST key property: `left subtree values < node.val < right subtree values`. This allows $O(h)$ search/insert/delete by binary-searching the tree.

BST Template: Pass Bounds Down

```
// Validate BST by passing min/max constraints down the tree
boolean validate(TreeNode node, long min, long max) {
    if (node == null) return true;
    if (node.val <= min || node.val >= max) return false; // ← bounds check
    return validate(node.left, min, node.val)           // left: tighten max
        && validate(node.right, node.val, max);         // right: tighten min
}

// Call with: validate(root, Long.MIN_VALUE, Long.MAX_VALUE)
// Use Long to handle Integer.MIN_VALUE / Integer.MAX_VALUE edge cases
```

BST Template: Inorder Traversal = Sorted Order

```
// BST inorder (left → node → right) visits nodes in ascending order
void inorder(TreeNode node) {
    if (node == null) return;
    inorder(node.left);
    // process node (kth smallest: increment counter, return when k)
    inorder(node.right);
}
```

BST Template: Binary Search on Tree

```
// Navigate BST like binary search: go left or right based on comparison
TreeNode search(TreeNode root, int target) {
    while (root != null) {
        if (root.val == target) {
            return root;
        } else if (root.val < target) {
            root = root.right;
        } else {
            root = root.left;
        }
    }
    return null;
}
```

Graph `graph.md`

`Queue<Integer> queue = new ArrayDeque<>()` throughout.

Six algorithms — each with a canonical template and representative problems.

Complexity Legend

BFS / DFS	$O(V + E)$	visit each vertex and edge once
Topo sort (Kahn)	$O(V + E)$	each node enqueued once, each edge relaxed once
Union-Find	$\sim O(n \cdot \alpha(n)) \approx O(n)$	α = inverse Ackermann, effectively constant
Dijkstra	$O((V + E) \log V)$	non-negative weights ONLY

Graph Representation

```
// Adjacency list (most common)
List<List<Integer>> graph = new ArrayList<>();
for (int i = 0; i < n; i++) graph.add(new ArrayList<>());
for (int[] edge : edges) {
    graph.get(edge[0]).add(edge[1]);
    graph.get(edge[1]).add(edge[0]);    // omit for directed
}

// Weighted adjacency list: int[] = {neighbor, weight}
List<List<int[]>> graph = new ArrayList<>();
for (int i = 0; i < n; i++) graph.add(new ArrayList<>());
for (int[] edge : edges)
    graph.get(edge[0]).add(new int[]{edge[1], edge[2]});

// Grid 4-directional neighbors
int[] dr = {1, -1, 0, 0};
int[] dc = {0, 0, 1, -1};
```

All Six Templates

```
// 1. BFS – shortest path / level order
// MENTAL MODEL: explore in rings of equal distance, so the first time you reach a node is the shortest
// WHEN: "fewest steps", "shortest path" on an unweighted graph
Queue<Integer> queue = new ArrayDeque<>();
boolean[] visited = new boolean[n];
queue.offer(start);
visited[start] = true;
while (!queue.isEmpty()) {
    int node = queue.poll();
    for (int neighbor : graph.get(node)) {
        if (!visited[neighbor]) {
            visited[neighbor] = true;
            queue.offer(neighbor);
        }
    }
}

// 2. TOPOLOGICAL SORT – Kahn's BFS
// MENTAL MODEL: peel off nodes with no remaining prerequisites; if any get stuck, a cycle exists.
// WHEN: "valid order", "dependencies/prerequisites", "detect cycle" on a directed graph
int[] inDegree = new int[n];
for (int[] edge : edges) inDegree[edge[1]]++;
Queue<Integer> queue = new ArrayDeque<>();
for (int i = 0; i < n; i++) if (inDegree[i] == 0) queue.offer(i);
List<Integer> order = new ArrayList<>();
while (!queue.isEmpty()) {
    int node = queue.poll();
    order.add(node);
    for (int neighbor : graph.get(node))
        if (--inDegree[neighbor] == 0) queue.offer(neighbor);
}
// order.size() == n → no cycle

// 3. UNION FIND
// MENTAL MODEL: each set points to one representative; merge sets by linking roots, query by finding r
// WHEN: "connected components", "is it already connected", "merge groups dynamically"
int[] parent, rank;
void init(int n) {
    parent = new int[n]; rank = new int[n];
    for (int i = 0; i < n; i++) parent[i] = i;
}
int find(int x) {
    if (parent[x] != x) parent[x] = find(parent[x]); // path compression
    return parent[x];
}
boolean union(int x, int y) {
    int px = find(x), py = find(y);
    if (px == py) return false;
    if (rank[px] < rank[py]) { int t = px; px = py; py = t; }
    parent[py] = px;
    if (rank[px] == rank[py]) rank[px]++;
    return true;
}

// 4. DIJKSTRA – shortest path (non-negative weights)
// MENTAL MODEL: greedily settle the closest unfinished node; non-negative weights guarantee it's final
// WHEN: "shortest/cheapest path" with non-negative weights
int[] dist = new int[n];
Arrays.fill(dist, Integer.MAX_VALUE);
dist[src] = 0;
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]); // min-heap on cost
```

```

pq.offer(new int[]{src, 0});
while (!pq.isEmpty()) {
    int[] current = pq.poll();
    int node = current[0], d = current[1];
    if (d > dist[node]) continue; // stale entry
    for (int[] nb : graph.get(node)) {
        int next = nb[0], w = nb[1];
        if (dist[node] + w < dist[next]) {
            dist[next] = dist[node] + w;
            pq.offer(new int[]{next, dist[next]});
        }
    }
}

// 5. BIPARTITE CHECK – BFS 2-coloring
// MENTAL MODEL: paint neighbors the opposite color; a neighbor that's already your color means an odd
// WHEN: "split into two groups", "2-colorable", "no edge within a group"
int[] color = new int[n];
Arrays.fill(color, -1);
Queue<Integer> queue = new ArrayDeque<>();
for (int start = 0; start < n; start++) {
    if (color[start] != -1) continue;
    color[start] = 0;
    queue.offer(start);
    while (!queue.isEmpty()) {
        int node = queue.poll();
        for (int neighbor : graph.get(node)) {
            if (color[neighbor] == -1) { color[neighbor] = 1 - color[node]; queue.offer(neighbor); }
            else if (color[neighbor] == color[node]) return false; // conflict
        }
    }
}
return true;

// 6. PRIM'S MST – greedy, always add cheapest edge to growing tree
// MENTAL MODEL: grow one tree outward, each step absorbing the cheapest edge that reaches a new node.
// WHEN: "connect all nodes at minimum total cost"
boolean[] inMST = new boolean[n];
int[] minEdge = new int[n];
Arrays.fill(minEdge, Integer.MAX_VALUE);
minEdge[0] = 0;
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]);
pq.offer(new int[]{0, 0});
int total = 0;
while (!pq.isEmpty()) {
    int[] current = pq.poll();
    int node = current[0], cost = current[1];
    if (inMST[node]) continue;
    inMST[node] = true;
    total += cost;
    for (int[] nb : graph.get(node))
        if (!inMST[nb[0]] && nb[1] < minEdge[nb[0]]) {
            minEdge[nb[0]] = nb[1];
            pq.offer(new int[]{nb[0], nb[1]});
        }
}

```

Single-Source vs Multi-Source

Single-source: one starting node, find distances to all others

Multi-source: seed queue with ALL starting nodes at once → BFS fans out from all simultaneously
→ avoids nested loops, handles "nearest X" questions in $O(n)$

Core idea: repeatedly remove nodes with `inDegree == 0` (no prerequisites). If all nodes removed → no cycle. Post-order in DFS = reverse of `order` here.

Template (always the same three methods):

```
int[] parent, rank;

void init(int n) {
    parent = new int[n];
    rank = new int[n];
    for (int i = 0; i < n; i++) parent[i] = i;
}

int find(int x) {
    if (parent[x] != x) parent[x] = find(parent[x]); // path compression
    return parent[x];
}

boolean union(int x, int y) { // returns false if already connected
    int px = find(x), py = find(y);
    if (px == py) return false;
    if (rank[px] < rank[py]) { int t = px; px = py; py = t; }
    parent[py] = px;
    if (rank[px] == rank[py]) rank[px]++;
    return true;
}
```

Dijkstra vs Bellman-Ford

	Dijkstra	Bellman-Ford
Weights:	Non-negative only	Any (handles negative)
Constraint:	—	k-hop limit (run k+1 rounds)
Time:	$O((V+E) \log V)$	$O(k \cdot E)$
When to use:	Standard shortest path	k-stop limit or negative weights
Key trick:	stale check: <code>d > dist[node]</code>	copy array each round (<code>temp[]</code>)

Algorithm Chooser

Signal in problem	Algorithm
Shortest path, unweighted / unit weight	BFS
Shortest path, non-negative weights	Dijkstra
Shortest path, k-hop limit	Bellman-Ford (k+1 rounds)
Shortest path, negative weights	Bellman-Ford (n-1 rounds)
Connected components, cycle detection	Union Find
Directed cycle / valid ordering	Topological Sort (Kahn's)
2-colorable graph	Bipartite BFS coloring
Connect all nodes minimum cost	Prim's / Kruskal's MST
"nearest X for all cells"	Multi-source BFS

Stale Entry Pattern (Dijkstra)

```
int[] current = pq.poll();
int node = current[0], d = current[1];
if (d > dist[node]) continue; // ← this node was already relaxed to a better distance
```

Without this check, stale entries in the heap cause re-processing and wrong results.

It replaces a `visited[]` array — simpler and correct because a better distance always wins.

Greedy `greedy.md`

Two interval templates plus non-interval greedy patterns.

Two Interval Templates

MERGE DIRECTION RULE (for merge-style problems like #56, where both keys work):

Sort by START → traverse FORWARD ($i = 0 \rightarrow n-1$), extend the END of last kept

Sort by END → traverse BACKWARD ($i = n-1 \rightarrow 0$), extend the START of last kept

These two are exact mirror images. See #56 for both written out side by side.

(NOTE: this mirror only applies to merging. Template 1 below sorts by end but still sweeps FORWARD — there the goal is counting, not merging.)

TEMPLATE 1 — Sort by END time, sweep forward, track end of last kept interval

Use when: maximizing count of non-overlapping intervals

Greedy insight: earliest-finishing interval leaves most room for future ones

TEMPLATE 2 — Sort by START time, sweep forward, merge when overlapping

Use when: merging/counting overlapping intervals

Greedy insight: processing in start order → each interval only needs to compare with the last merged result

```

// MENTAL MODEL: sort to expose a local greedy choice, then sweep once making the locally-best pick.
// WHEN: intervals + "max non-overlapping / min arrows / merge / min rooms", or jump/partition sweeps.
// TEMPLATE 1: Sort by end, compare start
Arrays.sort(intervals, (a, b) -> a[1] - b[1]); // sort by END time
int lastEnd = Integer.MIN_VALUE;
for (int[] interval : intervals) {
    if (interval[0] >= lastEnd) { // start >= lastEnd → no overlap → keep
        lastEnd = interval[1];
        // count++ or add to result
    } else {
        // overlap → skip (or count as removed)
    }
}

// TEMPLATE 2: Sort by start, compare end
Arrays.sort(intervals, (a, b) -> a[0] - b[0]); // sort by START time
List<int[]> result = new ArrayList<>();
for (int[] interval : intervals) {
    if (result.isEmpty() || result.get(result.size()-1)[1] < interval[0]) {
        result.add(interval); // no overlap → new interval
    } else {
        result.get(result.size()-1)[1] =
            Math.max(result.get(result.size()-1)[1], interval[1]); // overlap → extend end
    }
}

// TEMPLATE 3: Sort by start + min-heap of end times (multi-resource)
// USE WHEN counting simultaneous/overlapping resources – "minimum rooms", "max concurrent".
Arrays.sort(intervals, (a, b) -> a[0] - b[0]);
PriorityQueue<Integer> pq = new PriorityQueue<>(); // min-heap of active end times
for (int[] interval : intervals) {
    if (!pq.isEmpty() && pq.peek() <= interval[0])
        pq.poll(); // reuse: earliest-ending resource is free
    pq.offer(interval[1]); // occupy a resource until interval[1]
}
// pq.size() = resources needed

```

Sort-by-End vs Sort-by-Start+Heap

	Sort by End	Sort by Start + Heap
Goal:	count non-overlapping	count simultaneous active
Tracks:	lastEnd (scalar)	all active end times (heap)
Question answered:	max intervals I can keep	min rooms (resources) needed
Examples:	#435, #452, #646	#253 Meeting Rooms

Greedy Pattern Identifier

Signal	Template
"minimum removal to make non-overlapping"	Sort by end, keep non-overlapping (Template 1)
"minimum resources (arrows, groups)"	Sort by end, count groups (Template 1)
"merge overlapping intervals"	Sort by start, extend or add (Template 2)
"minimum rooms/machines"	Sort by start + min-heap (Template 3)
"can you reach the end?"	Track max reachable index
"minimum steps to reach the end?"	BFS layers via <code>currentEnd</code>
"partition string by last occurrence"	Track <code>last[]</code> array + extend <code>end</code>
"reconstruct order by two constraints"	Sort by one constraint, insert by the other
"can you complete the circuit / circular route?"	Track cumulative tank; reset start when tank < 0
"can person attend all meetings (no overlap)?"	Sort by start, compare adjacent intervals

Dynamic Programming `dynamic_programming.md`

Six template families. Every problem maps to one loop skeleton and one dp-state definition.

All Six Templates

```
// 1. LINEAR 1D – each cell depends on a fixed number of previous cells
// MENTAL MODEL: the answer at i is a fixed recipe over the last one or two answers.
// WHEN: "ways/cost to reach step i", "can't pick adjacent"
int[] dp = new int[n + 1];
dp[0] = base;
for (int i = 1; i <= n; i++)
    dp[i] = f(dp[i-1], dp[i-2], ...);

// 2A. LOOK-BACK 1D – dp[i] depends on all j < i
// MENTAL MODEL: to finish position i, scan every earlier position j and extend the best valid one.
// WHEN: "longest increasing/chain ending here", "can the prefix be segmented"
int[] dp = new int[n];
for (int i = 0; i < n; i++) {
    for (int j = 0; j < i; j++) {
        dp[i] = f(dp[i], dp[j]); // e.g., LIS, word break
    }
}

// 2B/2C. KNAPSACK – collapse the item dimension to 1D; loop direction picks reuse.
// 0/1 (each item once): j DESCENDING → reads previous row.
// unbounded (reusable): j ASCENDING → reads current row.
// Mnemonic: DESCENDING = Distinct, ASCENDING = Again. Full code + permutation
// variant + "why" in Part 2 – Knapsack DP.

// 3. 2D SEQUENCE – two strings/arrays; i indexes one, j indexes the other
// MENTAL MODEL: compare the last char of each prefix – match consumes both, mismatch drops one side.
// WHEN: "compare two strings/arrays" (LCS, edit distance, subsequence count)
int[][] dp = new int[m + 1][n + 1];
for (int i = 1; i <= m; i++) {
    for (int j = 1; j <= n; j++) {
        if (match(i, j)) {
            dp[i][j] = dp[i-1][j-1] + val;
        } else {
            dp[i][j] = combine(dp[i-1][j], dp[i][j-1]);
        }
    }
}

// 4. INTERVAL DP – i goes RIGHT to LEFT; j goes i+1 to end
// "shorter comes first": when computing dp[i][j], dp[i+1][*] is already done
// MENTAL MODEL: build answers for short ranges first, then combine them into longer ranges.
// WHEN: "palindrome substring/subseq", "merge/burst in a range", "split-point problems"
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base; // length-1 intervals
for (int i = n - 1; i >= 0; i--) { // ← right-to-left
    for (int j = i + 1; j < n; j++) {
        dp[i][j] = f(dp[i+1][j-1], dp[i+1][j], dp[i][j-1]);
    }
}

// 5. GRID DP – 4-directional neighbors
// MENTAL MODEL: each cell's answer accumulates from the cells you could have arrived from.
// WHEN: "paths / min-cost in a grid moving right+down", "largest square"
int[] dr = {1,-1,0,0}, dc = {0,0,1,-1}; // DOWN UP RIGHT LEFT
// Standard grid (dag, only right/down):
int[][] dp = new int[rows][cols];
for (int i = 0; i < rows; i++)
    for (int j = 0; j < cols; j++)
        dp[i][j] = grid[i][j] + combine(dp[i-1][j], dp[i][j-1]);

// 6. STATE MACHINE – named states updated simultaneously each step
```

```
// MENTAL MODEL: track the best value in each named situation; each day every state updates from yester
// WHEN: "buy/sell/hold", "limited transactions", "cooldown/fee"
int cash = 0, hold = -prices[0];
for (int i = 1; i < prices.length; i++) {
    cash = Math.max(cash, hold + prices[i]);
    hold = Math.max(hold, ??? - prices[i]);    // ??? depends on problem
}
```

Knapsack Decision Tree

```
Want count or min/max?
├─ Count (dp[j] += dp[j - num])
│   ├── 0/1 (each item once):    j descending → #416, #494
│   └─ Unbounded (reuse):       j ascending  → #518
│       └─ Ordered (permut):    target outer, nums inner → #377
└─ Min/Max (dp[j] = min/max(...))
    ├── 0/1 minimize:           j descending → #474 (maximize)
    └─ Unbounded minimize:     j ascending  → #322
```

Canonical Templates

```
// — 0/1 KNAPSACK — each item at most once
int[] dp = new int[target + 1];
dp[0] = 1;
for (int i = 0; i < nums.length; i++) {
    for (int j = target; j >= nums[i]; j--)    // DESCENDING: sees OLD dp (before item i)
        dp[j] += dp[j - nums[i]];
}

// — UNBOUNDED KNAPSACK — each item reusable
int[] dp = new int[target + 1];
dp[0] = 1;
for (int i = 0; i < nums.length; i++) {
    for (int j = nums[i]; j <= target; j++)    // ASCENDING: sees NEW dp (after item i already used)
        dp[j] += dp[j - nums[i]];
}

// — PERMUTATION COUNT — order matters (Combination Sum IV)
int[] dp = new int[target + 1];
dp[0] = 1;
for (int j = 1; j <= target; j++) {           // OUTER: target
    for (int num : nums) {                   // INNER: try all nums (not fixing order)
        if (j >= num) dp[j] += dp[j - num];
    }
}
}
```

Why descending for 0/1:

`dp[j - nums[i]]` is read **before** `dp[j]` is updated ($j > j - \text{nums}[i]$). So it always sees the dp state from before item `i` was considered — no item is used twice.

Why ascending for unbounded:

`dp[j - nums[i]]` was already updated in this same pass ($j - \text{nums}[i] < j$, processed earlier). This allows item `i` to be picked multiple times.

2D anchor: both loops are `dp[i][j] = dp[i-1][j] + dp[i-?][j-w]` collapsed to 1D — descending keeps `dp[j-w]` on the *previous* row (item `i` unused), ascending pulls it from the *current* row (item `i` reused).

Mnemonic: DESCENDING = **D**istinct (each item once), ASCENDING = **A**gain (reusable).

Flavor = seed + operator (same skeleton):

- count → `dp[0]=1, dp[j] += dp[j-w]`
- feasibility → `dp[0]=true, dp[j] = dp[j] || dp[j-w]`
- max-value → `dp[0]=0, dp[j] = Math.max(dp[j], dp[j-w] + v)`

Canonical Template

```
// i indexes string/array 1 (1-indexed), j indexes string/array 2
int[][] dp = new int[m + 1][n + 1];
// Init dp[0][*] and dp[*][0] based on problem
for (int i = 1; i <= m; i++) {
    for (int j = 1; j <= n; j++) {
        if (s1.charAt(i-1) == s2.charAt(j-1)) {
            dp[i][j] = dp[i-1][j-1] + 1; // characters match
        } else {
            dp[i][j] = combine(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]);
        }
    }
}
```

Canonical Template

Rule: `i` goes right-to-left (ensures shorter/inner intervals are computed first). `j` goes left-to-right from `i+1`. When computing `dp[i][j]`, all `dp[i'][j']` with `i' > i` or `j' < j` are already done.

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base_case; // single elements
for (int i = n - 1; i >= 0; i--) { // right-to-left
    for (int j = i + 1; j < n; j++) { // left-to-right from i+1
        // dp[i][j] can safely use dp[i+1][j], dp[i][j-1], dp[i+1][j-1]
        dp[i][j] = ...;
    }
}
```

Alternative (length-based): Equivalent, sometimes clearer for k-partition problems:

```
for (int len = 2; len <= n; len++) { // outer: length of interval
    for (int i = 0; i + len - 1 < n; i++) { // inner: left endpoint
        int j = i + len - 1;
        dp[i][j] = ...;
    }
}
```

Alternative (forward `i`, inner `j` descending): `i` is the right endpoint going left-to-right; `j` is the left endpoint sweeping back from `i-1`. Inner intervals (larger `j`, smaller `i`) are already filled.

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][i] = base_case; // single elements
for (int i = 0; i < n; i++) { // i = right endpoint, left-to-right
    for (int j = i - 1; j >= 0; j--) { // j = left endpoint, from i-1 down to 0
        // dp[i][j] can safely use dp[i-1][j], dp[i][j+1], dp[i-1][j+1]
        dp[i][j] = ...;
    }
}
```

Alternative (triangular fill, column base case): used when `dp[i][j]` depends only on the previous row/column (e.g. counting problems). Fill row by row with `j` from `0` to `i`.

```
int[][] dp = new int[n][n];
for (int i = 0; i < n; i++) dp[i][0] = 1; // base: dp[i][0] = 1; dp[0][j] = 0 for j > 0
for (int i = 1; i < n; i++) {
    for (int j = 1; j <= i; j++) { // j from 1 to i
        // dp[i][j] can safely use dp[i-1][j], dp[i][j-1], dp[i-1][j-1]
        dp[i][j] = ...;
    }
}
```

Canonical Template

```
int[] dr = {1,-1,0,0}, dc = {0,0,1,-1}; // DOWN UP RIGHT LEFT

// Standard grid (only move right/down - DAG, no cycle):
int[][] dp = new int[rows][cols];
// initialize first row and first column
for (int i = 0; i < rows; i++)
    for (int j = 0; j < cols; j++)
        dp[i][j] = grid[i][j] + f(dp[i-1][j], dp[i][j-1]);

// All-direction grid (memoized DFS for increasing/decreasing paths):
int[][] memo = new int[rows][cols];
for (int i = 0; i < rows; i++)
    for (int j = 0; j < cols; j++)
        dfs(grid, i, j, memo, dr, dc);
```

Canonical States

cash = max profit when NOT holding (free to buy or idle)
 hold = max profit when HOLDING stock (bought it at some cost)
 cooldown = max profit right after selling (can't buy today)

All Five Transitions

	buy	sell	re-buy condition
#121 (1 tx):	hold = max(hold, -price)		no re-buy (ignore cash)
#122 (unlimited):	hold = max(hold, cash - price)		re-buy with full cash
#123 (2 tx):	chain: buy2 uses sell1 cash		4 states chained
#309 (cooldown):	hold = max(hold, cooldown - price)		must wait 1 day
#714 (fee):	hold = max(hold, cash - price)		pay fee on sell

Stock Problems — Differences at a Glance

Problem	hold update	cash update	Extra state
#121	max(hold, -price)	max(cash, hold+price)	none
#122	max(hold, cash-price)	max(cash, hold+price)	none
#123	4 variables, chained	—	sell1 feeds buy2
#309	max(hold, cooldown-price)	max(cash, hold+price)	cooldown = prevCash
#714	max(hold, cash-price)	max(cash, hold+price-fee)	none

Template Chooser

Signal in problem	Template
<code>dp[i]</code> depends on <code>dp[i-1]</code> , <code>dp[i-2]</code>	Linear 1D — fixed lookback
<code>dp[i]</code> depends on all <code>dp[j]</code> where <code>j < i</code>	Look-back 1D — $O(n^2)$ inner loop
Each item used at most once, sum target	Knapsack 0/1 — <code>j</code> descending
Items reusable, sum target, order irrelevant	Knapsack Unbounded — <code>j</code> ascending
Items reusable, order matters (permutations)	Permutation count — target outer
Two strings/arrays, compare characters	2D Sequence DP
Palindrome, substring merge, balloon burst	Interval DP — <code>i</code> right-to-left
Grid, only right/down movement	Grid DP — cumulative from edges
Grid, all 4 directions, monotone constraint	DFS + memo (no cycle in DAG)
Buy/sell/hold decision changes day to day	State Machine — named states

Trie `trie.md`

When to Use Trie (vs HashMap/Array)

Signal	Use
Prefix queries / <code>startsWith</code> / autocomplete	Trie — shares prefixes, prefix lookup is $O(k)$
Wildcard match (<code>.</code> matches any char)	Trie — DFS branches over children
Prune a search space by shared prefixes (e.g. Word Search II)	Trie — dead branches cut early
Exact-key lookup only (no prefix logic)	HashMap — $O(k)$ hash is simpler and faster, no node overhead

Canonical Template

```
// MENTAL MODEL: every node is a shared prefix; walking down spells a word, so common prefixes are stor
// WHEN: "prefix / startsWith / autocomplete", "wildcard match", "prune a search by shared prefixes"

// — TRIE NODE —
// Array vs Map TrieNode mental model:
// array = O(1) per char, fixed 26 letters, wastes space when sparse
// HashMap = variable alphabet, smaller for sparse, hashing overhead
class TrieNode {
    TrieNode[] children = new TrieNode[26]; // array-based: O(1) access, O(26) per node
    boolean isEnd = false;
}
// Alternative: Map-based (for non-lowercase or variable alphabets)
class TrieNode {
    Map<Character, TrieNode> children = new HashMap<>();
    boolean isEnd = false;
}

// — INSERT —
void insert(TrieNode root, String word) {
    TrieNode node = root;
    for (char c : word.toCharArray()) {
        int idx = c - 'a';
        if (node.children[idx] == null) node.children[idx] = new TrieNode();
        node = node.children[idx];
    }
    node.isEnd = true;
}

// — SEARCH (exact match) —
boolean search(TrieNode root, String word) {
    TrieNode node = root;
    for (char c : word.toCharArray()) {
        int idx = c - 'a';
        if (node.children[idx] == null) return false;
        node = node.children[idx];
    }
    return node.isEnd; // must end at a marked word boundary
}

// — STARTS WITH (prefix match) —
boolean startsWith(TrieNode root, String prefix) {
    TrieNode node = root;
    for (char c : prefix.toCharArray()) {
        int idx = c - 'a';
        if (node.children[idx] == null) return false;
        node = node.children[idx];
    }
    return true; // ← difference from search: don't check node.isEnd
}
```

Variations

```
// VARIATION 1: DFS for wildcard '.' (Add and Search Words)
// When char is '.', try all 26 children recursively instead of following one path.
boolean dfs(TrieNode node, String word, int i) {
    if (i == word.length()) return node.isEnd;
    char c = word.charAt(i);
    if (c == '.') {
        for (TrieNode child : node.children) // ← VARIATION: try all children
            if (child != null && dfs(child, word, i+1)) return true;
        return false;
    }
    int idx = c - 'a';
    if (node.children[idx] == null) return false;
    return dfs(node.children[idx], word, i + 1);
}

// VARIATION 2: early-exit on isEnd (Replace Words – find shortest root)
String findRoot(TrieNode root, String word) {
    TrieNode node = root;
    for (int i = 0; i < word.length(); i++) {
        int idx = word.charAt(i) - 'a';
        if (node.children[idx] == null) return word; // no root found → return original
        node = node.children[idx];
        if (node.isEnd) return word.substring(0, i+1); // ← VARIATION: return at first root hit
    }
    return word;
}

// VARIATION 3: store words at each node (Search Suggestions – top-3 per prefix)
// During insert, each node on the path caches the word (up to 3).
// Products must be inserted in sorted order → first 3 are lexicographically smallest.
void insert(TrieNode root, String word) {
    TrieNode node = root;
    for (char c : word.toCharArray()) {
        int idx = c - 'a';
        if (node.children[idx] == null) node.children[idx] = new TrieNode();
        node = node.children[idx];
        if (node.words.size() < 3) node.words.add(word); // ← VARIATION: cache at every node
    }
    node.isEnd = true;
}

// VARIATION 4: Trie + grid DFS pruning (Word Search II)
// Build Trie from word list. DFS the grid; at each cell, check if current path
// follows a Trie path. Prune entire branch if no Trie node exists.
void dfs(char[][] board, int i, int j, TrieNode node, StringBuilder path, Set<String> result) {
    if (i < 0 || i >= board.length || j < 0 || j >= board[0].length) return;
    char c = board[i][j];
    if (c == '#' || node.children[c - 'a'] == null) return; // ← VARIATION: Trie prune
    node = node.children[c - 'a'];
    path.append(c);
    if (node.isEnd) result.add(path.toString());
    board[i][j] = '#';
    dfs(board, i+1, j, node, path, result);
    dfs(board, i-1, j, node, path, result);
    dfs(board, i, j+1, node, path, result);
    dfs(board, i, j-1, node, path, result);
    board[i][j] = c;
    path.deleteCharAt(path.length() - 1);
}
```

search vs startsWith — One Line Difference

```
search:      return node.isEnd;    // must land on a word boundary
startsWith:  return true;          // any node reached = valid prefix
```

Trie vs HashMap vs Sorting — When to Use Trie

Operation	HashMap	Sorting	Trie
Exact key lookup	$O(k)$ ✓	$O(\log n \cdot k)$	$O(k)$
Prefix check	$O(k)$ per key	$O(\log n)$	$O(k)$ ✓
All words with prefix	$O(n \cdot k)$	$O(\log n + \text{result})$	$O(k + \text{result})$ ✓
Wildcard match	Not supported	Not supported	$O(26^k)$ with DFS ✓
Space	$O(n \cdot k)$	$O(n \cdot k)$	$O(n \cdot k)$ alphabet-dependent

Use Trie when: prefix queries, wildcard search, or pruning a search space by shared prefixes (Word Search II).

Insert → Search — Structural Parallel

```
Both traverse the same path. Insert creates nodes; search reads them.
The only decision point after traversal:

search:      return node.isEnd      ← complete word?
startsWith:  return true             ← any prefix is fine
findRoot:    return early if isEnd   ← want the shallowest word boundary
cache words: node.words.add(word)    ← accumulate at every node during insert
```

Bit Manipulation `bit_manipulation.md`

Canonical Template

```
// MENTAL MODEL: bits are a set/counter – XOR cancels duplicates, AND/OR/shift edit individual bits.
// WHEN: "appears once/odd among pairs", "count set bits", "no shared letters", "power of 2/4".
// — XOR CANCEL — pairs cancel; lone element survives
int result = 0;
for (int num : nums) result ^= num;

// — BIT OPERATIONS —
boolean isSet = ((n >> i) & 1) == 1;    // check if bit i is set
int set      = n | (1 << i);           // set bit i
int clear    = n & ~(1 << i);          // clear bit i
int toggle   = n ^ (1 << i);           // toggle bit i
int lowest   = n & (-n);                // isolate lowest set bit
boolean isPow2 = n > 0 && (n & (n-1)) == 0;

// — COUNT SET BITS — Brian Kernighan: each iteration removes one set bit
// WHY n & (n-1) clears the lowest set bit:
//   n      = ...1000 (lowest set bit at this position)
//   n-1    = ...0111 (borrow flips that bit to 0 and all lower bits to 1)
//   n&(n-1)=...0000 (AND keeps higher bits, wipes the lowest set bit + below)
int count = 0;
while (n != 0) { count++; n &= (n - 1); } // n & (n-1) clears lowest set bit

// — BITMASK AS SET — 26-bit integer represents presence of 26 letters
int mask = 0;
for (char c : word.toCharArray()) mask |= (1 << (c - 'a'));
// check no overlap between two words:
if ((mask[i] & mask[j]) == 0) { /* no shared letter */ }
```

Variations

```
// VARIATION 1: mod-3 state machine (Single Number II)
// MENTAL MODEL: extend XOR from mod-2 (cancel pairs) to mod-3 (cancel triples);
//               ones/twos track bits seen 1 or 2 times mod 3.
// Instead of XOR canceling pairs (mod 2), cancel triples (mod 3)
// ones = bits seen exactly 1 mod 3 times; twos = bits seen exactly 2 mod 3 times
int ones = 0, twos = 0;
for (int num : nums) {
    ones = (ones ^ num) & ~twos;    // ← add to ones if not already in twos
    twos = (twos ^ num) & ~ones;    // ← add to twos if not already in ones
}
// After all: ones holds the element appearing once (0 mod 3 remainder = survive)

// VARIATION 2: split XOR into two groups (Single Number III)
// Two unique elements a, b. XOR of all = a ^ b.
// Use lowest set bit of (a^b) to split nums into two groups → each group has one unique.
int xor = 0;
for (int num : nums) xor ^= num;    // xor = a ^ b
int diff = xor & (-xor);            // ← lowest set bit that differs between a and b
int a = 0;
for (int num : nums)
    if ((num & diff) != 0) a ^= num; // ← XOR one group → one unique element
int b = xor ^ a;                    // ← the other unique element

// VARIATION 3: DP relation (Counting Bits)
// dp[i] = dp[i/2] + last bit of i
// i >> 1 = i without last bit (same or fewer ones); last bit = i & 1
int[] dp = new int[n + 1];
for (int i = 1; i <= n; i++)
    dp[i] = dp[i >> 1] + (i & 1);    // ← right-shift halves the problem

// VARIATION 4: common prefix (Bitwise AND of Range)
// AND of any range [left, right]: bits that differ between left and right get cleared.
// Right-shift both until equal → find shared high-bit prefix → shift back.
int shift = 0;
while (left != right) { left >>= 1; right >>= 1; shift++; }
return left << shift;                // ← shift back to original magnitude

// VARIATION 5: carry loop (Sum of Two Integers)
// XOR gives bit sum without carry; AND shifted left gives carry.
// Repeat until no carry remains.
int add(int a, int b) {
    while (b != 0) {
        int carry = a & b;           // ← carry bits
        a = a ^ b;                   // ← partial sum (no carry)
        b = carry << 1;              // ← carry propagated one position left
    }
    return a;
}
```

Bit Trick Cheat Sheet

```
n & (n-1)           // clear lowest set bit → count bits, check power of 2
n & (-n)            // isolate lowest set bit → split into groups
n ^ n = 0           // same value cancels → XOR pairs
n ^ 0 = n           // identity → XOR with index
(n >> i) & 1        // check bit i
n | (1 << i)        // set bit i
n & ~(1 << i)       // clear bit i
n ^ (1 << i)        // toggle bit i
i >> 1              // divide by 2 (fast)
i & 1               // last bit (0 = even, 1 = odd)

// --- Divisibility by powers of two ---
(n & 1) == 0        // divisible by 2 (even) ← LSB is the 2^0 place
(n & 1) == 1        // NOT divisible by 2 (odd)
(n & ((1 << k) - 1)) // remainder of n / 2^k ; == 0 means divisible by 2^k
(n & 3) == 0        // divisible by 4 (k = 2)
(n & 7) == 0        // divisible by 8 (k = 3)
// Note: n & 1 is safe for NEGATIVE numbers (two's complement) where
// n % 2 returns -1 for odd negatives. Prefer & 1 for parity.
```

Pattern Chooser

Signal	Technique	Time
"appears odd/once, rest appear even/twice"	XOR all	O(n)
"appears once, rest appear 3 times"	mod-3 state machine (ones , twos)	O(n)
"two unique elements"	XOR → split by xor & (-xor)	O(n)
Count set bits	Brian Kernighan n &= (n-1)	O(k) k = set bits
Count bits for all 0..n	DP: dp[i] = dp[i>>1] + (i&1)	O(n)
AND of a range	Right-shift to find common prefix	O(log n)
Add without +	XOR + carry loop	O(1) (32 iters)
"no shared characters between two strings"	26-bit bitmask, check AND == 0	O(n²) pairwise
"is n a power of two?"	n > 0 && (n & (n-1)) == 0	O(1)
"is n a power of four?"	Power of 2 AND set bit at even position: (n & 0xAAAAAAAA) == 0	O(1)
"find duplicate, no extra space, no modification"	Floyd's cycle detection on array-as-linked-list	O(n)

Design `design.md`

Canonical Template — HashMap + Doubly Linked List (LRU)

```
// MENTAL MODEL: HashMap gives O(1) lookup; the doubly linked list orders by recency so the LRU is alwa
// WHEN: "O(1) get/put with eviction by recency or frequency"

// CacheNode for doubly linked list (named CacheNode to distinguish from ListNode contexts)
class CacheNode {
    int key, value;
    CacheNode previous, next;
    CacheNode(int key, int value) { this.key = key; this.value = value; }
}

// — SENTINEL NODES — head (MRU side) ↔ ... ↔ tail (LRU side)
CacheNode head = new CacheNode(0, 0), tail = new CacheNode(0, 0);
// Initialize: head <-> tail
head.next = tail; tail.previous = head;

// — REMOVE NODE —
void remove(CacheNode node) {
    node.previous.next = node.next;
    node.next.previous = node.previous;
}

// — INSERT AFTER HEAD (= mark as MRU) —
void insertAfterHead(CacheNode node) {
    node.next = head.next;
    node.previous = head;
    head.next.previous = node;
    head.next = node;
}

// — LRU EVICTION — tail.previous is LRU
CacheNode lru = tail.previous;
remove(lru);
map.remove(lru.key);
```

Variations

```
// VARIATION 1: LFU — adds a frequency dimension
// Need: key->value, key->freq, freq->orderedSet<key>, and minFreq
// When a key is accessed: freq++, move from freqToKeys[oldFreq] to freqToKeys[newFreq]
// On eviction: remove first element from freqToKeys[minFreq]
// LinkedHashMap preserves insertion order → oldest at front for same frequency

// VARIATION 2: Insert Delete GetRandom — use array for O(1) random
// ArrayList stores values; HashMap stores val->index in array
// On remove: swap target with last element, update map, remove last
// On getRandom: return list.get(random.nextInt(list.size()))
```

LRU vs LFU vs RandomizedSet — Pattern Chooser

Signal	Design
"evict least recently used"	LRU: HashMap + doubly linked list, move to head on access
"evict least frequently used"	LFU: 3 HashMaps + LinkedHashMap + minFreq tracking
"O(1) insert/delete/random"	RandomizedSet: HashMap + ArrayList, swap-with-last on delete

Key Invariants

LRU: `head.next = MRU`, `tail.previous = LRU`

LFU: `minFreq` always points to the smallest existing frequency bucket

`LinkedHashSet` preserves insertion order → `iterator().next()` = LRU within bucket

`RandomizedSet`: `list[map[val]] == val` at all times (after swap, update the map for the swapped key)

Math `math.md`

Canonical Templates

```
// MENTAL MODEL: number = sequence of digits in some base; iterate by repeatedly
//                  peeling the last digit (% base) and shifting (/ base).
// WHEN: "reverse digits", "x^n", "base conversion", "count primes", "nth divisible-by".
```

```
// — DIGIT EXTRACTION LOOP — peel digits right-to-left
```

```
while (n != 0) {
    int digit = n % 10; // last digit
    n /= 10;           // drop last digit
    // ... use digit ...
}
```

```
// — FAST POWER (exponentiation by squaring) —
```

```
// WHY:  $x^n$  needs only  $O(\log n)$  multiplications by squaring the base
```

```
// and consuming one exponent bit per step.
```

```
//  $x^{13} = x^8 * x^4 * x^1$  (13 = 1101 in binary)
```

```
double fastPow(double x, long n) {
    double result = 1;
    while (n > 0) {
        if ((n & 1) == 1) result *= x; // bit set → include this power of x
        x *= x;                       // square the base for the next bit
        n >>= 1;                      // consume one exponent bit
    }
    return result;
}
```

```
// — BASE CONVERSION — generic base b, 0-indexed
```

```
// number → digits: peel with % b, shift with / b, collect, then reverse
```

```
StringBuilder builder = new StringBuilder();
```

```
while (num > 0) {
    builder.append(num % b);
    num /= b;
}
```

```
builder.reverse();
```

```
// digits → number: Horner's rule
```

```
int value = 0;
for (int digit : digits) value = value * b + digit;
```

```
// — SIEVE OF ERATOSTHENES — mark composites up to n
```

```
// WHY start at  $i*i$ : any multiple  $k*i$  with  $k < i$  was already marked by  $k$ .
```

```
boolean[] composite = new boolean[n];
```

```
for (int i = 2; i * i < n; i++) {
    if (composite[i]) continue;
    for (int j = i * i; j < n; j += i) // start at  $i*i$ , step by i
        composite[j] = true;
}
```

Math Cheat Sheet

```
n % 10          // last decimal digit
n /= 10         // drop last digit
result * 10 + digit // push a digit (build number left-to-right)
(n & 1) == 1     // odd
(n & 1) == 0     // even
n >>= 1         // halve (consume one bit)
x *= x          // square base in fast power
c - '0'         // char → digit value
c - 'A' + 1     // Excel letter → 1-indexed value
(char)('A' + d) // 0..25 → 'A'..'Z'
i * i           // sieve marking start point
x / gcd(x,y) * y // LCM without overflow (divide first)
```

Pattern Chooser

Signal	Technique	Time
"reverse the digits of a number"	Digit pop (%10) / push (*10) loop	O(log n)
"is the number a palindrome?"	Reverse only half, compare halves	O(log n)
"compute x^n / a^b efficiently"	Exponentiation by squaring (binary exp)	O(log n)
"x^huge mod m" / exponent given as array	Digit-by-digit modular fast power	O(log n)
"add two big numbers as strings"	Right-to-left add with carry	O(n)
"convert column number ↔ Excel title"	Base-26 conversion, 1-indexed	O(log n) / O(L)
"convert between number and base-b string"	Horner (parse) / peel-and-reverse (build)	O(L)
"count primes below n"	Sieve of Eratosthenes, mark from i*i	O(n log log n)
"nth number divisible by a/b/c"	Binary search on answer + inclusion-exclusion (LCM)	O(log range)
"check if n is power of 2/4"	Bit trick n & (n-1) == 0 (see bit_manipulation)	O(1)